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Hedging in Academic Writing: Some Theoretical Problems

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Abstract—Recent studies of hedging in academic writing have argued for the inclusion of hedging in EAP syllabi but have not, unfortunately, worked from a common understanding of the concept. This paper reviews and evaluates some of the different ways in which the term *hedge* has been understood and defined in the literature. Although the use of hedges as a politeness strategy offers the fullest functional account of hedging in academic writing, there has been a tendency to extend the reference of *hedge* to politeness-related features of academic writing, such as impersonal constructions, the use of the passive, and lexis-projecting emotions. It is suggested that *hedge* is more usefully reserved for expressions of epistemic modality, or markers of provisionality, as attached to new knowledge claims. It is further argued that it is not possible to designate certain kinds of lexis as epistemically modal and that authors can only be held responsible for modalizing, or hedging, their own propositions. A new definition of *hedge*, closely related to the ordinary definition, is suggested, together with a taxonomy of the hedges which would fit this definition so far identified in academic writing. © 1997 The American University. Published by Elsevier Science Ltd

Introduction

Most recent researchers on the subject are keen for hedging to be included in EAP programmes (Hyland 1994; Salager-Meyer 1994; Skelton 1988a). However, as Hyland's recent review of the literature puts it, "it is important to acknowledge differences between what is actually being measured" (p. 243). Unless or until a definition and a clear description can be achieved there seems little hope of studying or teaching the phenomenon consistently.

Hedge as a linguistic concept was introduced by G. Lakoff (1972). As this concept is far from being clear, however, I would like to begin by reconsidering the ordinary use of *hedge*. Without having to concern them-

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selves with the precise *forms* hedging takes, the compilers of the following dictionary definitions boldly address the *function* of hedging:

> 9. *intr.* To go aside from the straight way; to shift, shuffle, dodge; to trim; to avoid committing oneself irrevocably; to leave open a way of retreat or escape [Oxford English Dictionary (OED), Murray 1933: *V*, 188].
>
> 3. If you *hedge* or if you *hedge* a problem or question, you avoid answering the question or committing yourself to a particular action or decision (Collins COBUILD English Language Dictionary, Sinclair 1987: 677).

Hedge sounds as if it has a metaphorical origin and it is interesting to note that most of the OED definition is serviceable only metaphorically—we wouldn't say "Although the road was clear I hedged", or "He tried to hit me but I hedged". *Hedge* is presumably a buried metaphor, related to the "ARGUMENT IS WAR" family of metaphors identified by Lakoff and Johnson (1980). The OED definition evokes guerrilla-style tactics: no fixed defensive positions, concealment, camouflage, retaining the option of withdrawal. The COBUILD definition echoes the non-metaphorical component of the OED entry, the avoidance of personal commitment.

Functions and Forms of Hedges

If we bear its ordinary meaning in mind, G. Lakoff's (1972) conversion of *hedge* into a noun and his use of the word to describe a linguistic phenomenon can immediately be seen as problematic. Firstly there is the question of function. Is a certain word or phrase, for example "suggest", always a hedge? If not, how can we tell those instances where it is serving as a hedge from those where it is not? Some kind of analysis is required. Lakoff's definition of *hedges* as "words whose job is to make things fuzzy or less fuzzy" has been described as "an extension of the colloquial sense" (Brown & Levinson 1987: 145). However, words which make things less fuzzy seem likely to increase rather than avoid the utterer's commitment; the new linguistic definition seems therefore actually to contradict rather than "extend" the ordinary use of *hedge*. The function of ordinary (dare one say *common* or *garden*?) hedging is clear; the functions performed by Lakoffian hedges are possibly myriad. What are these functions?

Secondly, there is the question of form: Lakoff's nominalisation of *hedge* also suggests that *hedges* are a discrete set of linguistic items. If this is the case we are still a long way from developing a taxonomy of such items. What are the various kinds of hedge?

All subsequent researchers have been obliged to offer their own answers to these basic questions about function and form. As we shall see, little agreement has so far been reached.

Fuzziness: Approximators and Shields

Addressing the question of function, R. Lakoff (1972) introduced the subcategory of *performative* hedges, which modify the illocutionary force of

the speech act they accompany. The area of performative hedging has been studied in some detail in relation to politeness and acts such as requesting, ordering, inviting, declining requests, etc. (*cf.* Brown & Levinson 1987). However, much of this is not directly relevant to academic writing, in which, generally speaking, the main speech act performed is that of stating a proposition.

Building on the work of the Lakoffs, Prince et al. (1982) attempted to address the function of hedges in an empirical study of spoken medical discourse. They counted the number of words or phrases in their corpus which made things "fuzzier", and analysed each item as falling into one of two main categories: *approximators*, which introduce fuzziness "within the propositional content proper", and *shields*, which introduce fuzziness "in the relationship between the propositional content and the speaker" (p. 86). On these grounds they designated *approximators* as a semantic phenomenon and *shields* as a pragmatic one and judged the two classes of hedge to "have little in common". The function of *approximators* is either to adapt a term to a non-prototypical situation (e.g. "sort of vertical") or to indicate that a term is a rounded-off representation of some figure (e.g. "about ten fifty over five fifty"). *Shields*, by contrast, serve as a "a linguistic reflex of a marked commitment on the part of the speaker to the truth of the proposition that s/he is conveying" (p. 94). Within *shields* they identified two further subclasses: *plausibility* shields "which involve something related to doubt" (e.g. "I don't see that you have anything to lose by..."); and *attribution* shields which "attribute the belief in question to someone other than the speaker" (e.g. "according to her estimates") (p. 89).

In the conversations Prince et al. analysed, the social goal was making an accurate diagnosis and their analysis of functions is coloured by the nature of their corpus. So, for example, *attribution shields* were a means of according status to reported information (e.g. within the hierarchy "physician, nurse, parent") rather than a means of acknowledging scholarly debt or deliberately pointing up limitations of current knowledge in order to "create a research space" (Swales 1981, 1990). Nevertheless, their characterization of the overall function of *shields* is clear and economical, and anticipates later descriptions of hedges as expressing epistemic modality.

Commentative Language

Skelton (1988a) objects to the "unwelcome connotations" which the ordinary, "pejorative", meaning of *hedge* attracts and proposes the abandonment of the term, in favour of a distinction between *proposition* and *comment* (p. 38). Under this scheme hedges would be designated *commentative* language, which serves the function of *modulating* propositions: "It is by means of hedging that a user distinguishes between what s/he says and what s/he thinks about what she says" (p. 38).

However, while I agree with Skelton's general point that hedges "are a resource not a problem" (p. 39) I fear that if, in an effort to confer moral

respectability, we abandon the term *hedge* altogether, we abandon a useful label; the proposed substitute *comment* has far too large a scope, and like other euphemisms seems likely to lead to ambiguity. There are kinds of comments on propositions which express attitudes other than uncertainty; e.g.

1. It's raining, unfortunately.
2. I'm afraid it's raining.

These are formally similar but functionally quite different from:

1a. It's raining, probably.
2a. I feel sure it's raining.

All four statements involve what Skelton proposes to call *commentative* language but the speaker's comments in (1a) and (2a) do more than simply express attitude. They express the speaker's level of commitment to the truth of his or her proposition. In other words, hedging language seems to be a subset of commentative language. Skelton himself acknowledges that all language choices could be interpreted as comment and hence eschews a selection principle: he limits his empirical work to what he admits is an arbitrary selection of those linguistic occurrences which can be easily quantified ("Type 1 comments"; Skelton 1988b: 99).

Skelton (1988a: 40) also discusses commentative language partly in terms of *evaluativity*, as contrasted with *propositionality*, or *factuality*; however, he fails to distinguish between the factual language of science textbooks, which present a body of knowledge agreed on by the discourse community, and the conventionally much more tentative, and therefore, in his own terms, *evaluative* language of research writing. This distinction is crucial for budding academic writers: as students they may well have been expected to present and articulate items from the body of agreed knowledge without evaluation. The use of a propositional, or textbook-like, style will be acceptable. If, however, they aspire to present their own research findings to their peers—other researchers in the same field—such a style would be considered inappropriate and immodest. Academic writers need to make a clear distinction between propositions already shared by the discourse community, which have the status of facts, and propositions to be evaluated by the discourse community, which only have the status of claims. Evaluative or tentative language is one of the signs by which claims may be distinguished from facts; as Myers (1989: 13) argues, "a sentence that looks like a claim but has no hedging is probably not a statement of new knowledge".

Modesty and Politeness

The relationship between hedging and modesty seems to require clarification. Salager-Meyer (1994: 152) argues that "hedges are first and foremost the product of a mental attitude which looks for proto-typical linguistic

forms". She doesn't attempt a definition of *hedge* but characterizes the "functional concept" she used for her research as embracing three dimensions:

1. that of purposive fuzziness and vagueness (threat-minimizing strategy);
2. that which reflects the authors' modesty for their achievements and avoidance of personal involvement; and
3. that related to the impossibility or unwillingness of reaching absolute accuracy and of quantifying all the phenomena under observation (p. 153).

However, in her subsequent taxonomy of hedges, Salager-Meyer includes expressions of "authors' personal doubt and direct involvement". In fact, far from *avoiding* personal involvement, both modesty and the minimizing of threat to the discourse community (dimension 1) may be achieved by *projecting* the authors' involvement. The following claim is an example:

> As far as I can see, Carroll's conclusion still stands. (Lado 1986: 138)

Were Lado to avoid personal involvement, e.g.

> Carroll's conclusion still stands.

It is clear that he would sound both immodest and threatening in relation to the discourse community.

Rather than seeing hedging as a reflex of personal qualities such as "attitude" and "modesty", which sound like matters of taste, hedges are perhaps better understood as a product of social forces. Myers (1989) offers a rationale for hedges along with several other conventions in academic writing by applying Brown and Levinson's (1987) anthropological model of politeness. According to their model, politeness is "a strictly formal system of rational practical reasoning" (p. 58). Myers demonstrates that the same social variables which affect outcomes in everyday social interactions—social distance, power difference, and rank of imposition—exist in academic writing and lead to similar outcomes. The social ends of academic writers might be crudely summarized as "making a name for themselves" or in Swales' (1990) terminology *establishing* and *filling a niche*; these ends necessarily involve imposing on the face of others and simultaneously saving their own face (or defending themselves from threat). Such ends oblige academic writers to adopt the same linguistic strategies as are used in other social interactions. As Myers argues, any academic knowledge claim is a threat, or Face Threatening Act (FTA), to other researchers in the field because it infringes on their "freedom to act". The hedge discussed previously, therefore, "As far as I can see" mitigates the claim being made in that "the readers are still allowed to judge for themselves" (Myers 1989: 16). Myers accounts for hedging in academic writing as one of a range of politeness strategies:

> Hedging is a politeness strategy when it marks a claim, or any other statement, as being provisional, pending acceptance in the literature, acceptance by the community—in other words, acceptance by the readers. (p. 12.)

It seems worth noting that the underlying military metaphor in politeness theory—"threat-minimizing strategy"—matches well with the ordinary use of *hedge*. Another hedge-related metaphor which springs to mind is that of "beating about the bush", an activity whereby the speaker's social ends are camouflaged out of politeness. However, as Myers shows, hedging is only one of the politeness strategies classified by Brown and Levinson (1987) which is carried over to academic writing; others include, for example, "7. Impersonalise Speaker and Hearer", "8. State the FTA as a general rule", and "9. Nominalise" (p. 131). One strategy is to avoid making the FTA altogether; it is clear, however, that this can never be the sole strategy for a budding researcher; the social role requires that claims be made. The tension in politeness theory between the need to commit FTAs and the need to mitigate them, parallels the offensive/defensive tension expressed in the metaphor of hedging. Researchers need to advance claims but, out of politeness, which in this context means deference to the discourse community, may not, unilaterally as it were, commit themselves to these claims.

Salager-Meyer's (1994) taxonomy of hedges includes *emotionally charged intensifiers* (e.g. "particularly encouraging"). The reason for including this category of expressions as hedges is not accounted for within her functional concept of hedging. Once again politeness theory seems to offer the most lucid explanation of the function of such items, this time in terms of *positive* politeness strategy—that is politeness addressed to the positive face of the hearer. Such lexis as "particularly encouraging" shows "solidarity with the (discourse) community by exhibiting responses that assume shared knowledge and desires", showing "identification with a common goal, rather than the response or desires of an individual" (Myers 1989: 8). However, although hedges can be politeness strategies, this is not to say that all politeness strategies are hedges. If we designate, as Salager-Meyer effectively does, other politeness strategies as hedges, we need to make clear the basis for doing so.

Epistemic Modality

Leaving aside politeness theory, a slightly different rationale behind hedging in academic writing is emphasized by Hyland (1994: 240), who argues that "academics are crucially concerned with varieties of cognition, and cognition is inevitably 'hedged'". He identifies hedging with *epistemic modality* as defined by Lyons:

> Any utterance in which the speaker explicitly qualifies his commitment to the truth of the proposition expressed by the sentence he utters... is an epistemically modal or modalised sentence. (Lyons 1977: 797)

This concept can be seen as compatible with the *provisionality* of knowledge claims described by Myers. How these concepts are practically related to hedging forms is a critical issue.

Taxonomies of Hedges

Arising from the lack of a satisfactory definition of a hedge, researchers into hedges face two main kinds of problems. The first is that forms which have been identified as hedges have other functions too—although the absence of a satisfactory definition of the function(s) of hedges makes the precise "otherness" difficult to determine. As Hyland observes, for example, counting modal verbs is unhelpful because of "the degree of indeterminacy between the root and epistemic meanings of modal verbs" (Hyland 1994: 243). Salager-Meyer tackles this kind of problem in her survey by attempting to consider both formal and functional criteria: to assess the latter in her medical research corpus she employed the services of a specialist informant.

The second problem is that of possibly overlooking hedges which appear in forms which have not yet been identified as hedges. Brown and Levinson (1987: 146) argue that hedging is "a productive linguistic device" and "can be achieved in an indefinite number of surface forms". Hyland (1994: 243) refers to hedges taking "unpredictable forms, for example, by referring to the uncertain status of information". Salager-Meyer (1994: 154) gets around this problem by considering only "linguistic expressions *commonly regarded* as hedges" [italics added], which is hardly satisfactory theoretically.

Let us briefly review those forms researchers have chosen to regard as hedges. Unfortunately, the criteria Prince et al. (1982) used for identifying fuzziness are not available. Skelton's "Type 1 comments" mostly turn out be related to uncertainty: copulas other than *be*, modal auxiliaries, lexical verbs such as "believing", "arguing". A fourth class, however, is defined as "adjectivals or adverbials introduced by "It is", "This is", "There is", or which are sentence or clause initial and immediately followed by a comma" (Skelton 1988b: 100–101). These can be related to possibility or certainty but may also relate to significance or interest.

Myers (1989: 13), who is only concerned with hedges as realizations of politeness strategies, lists modal conditional verbs and modifiers but then adds with abandon "any device suggesting alternatives—anything but a statement with a form of *to be* that such and such is the case". Hyland (1994: 240) omits "approximators" but includes as well as epistemically modal expressions "IF-clauses, question forms, passivisation, impersonal phrases, and time reference". Unfortunately, Hyland gives no examples; let us consider the following example of an impersonal and passive construction:

> In constructing such measures (for both native and non-native speakers) the basic assumption has been that reading is made up of a number of skills. (Jafarpur 1987: 195)

Myers would follow Brown and Levinson in regarding this kind of *impersonality* as a negative politeness strategy (number 7). The FTA of criticising the "assumers" in the discourse community—"assumption" is after all unscientific—is mitigated by Jafarpur phrasing his observation "as if the addressee were other than H (hearer) or only inclusive of H" (Brown & Levinson 1987: 190). The sentence's nominalization is itself another strategy (number 9, p. 207–208): Brown and Levinson claim that "the more nouny an expression, the more removed an actor is from doing or feeling or being something" and therefore the less "dangerous" the FTA seems to be. As far as Hyland's listing of *passivization* as a hedge is concerned, Brown and Levinson also suggest that passives have "roughly adjectival status"; that is they come between the informality of verbs and the formality of nouns, in terms of politeness, "on the continuum from verb through adjective". However, it is again unclear on what basis *impersonality* and *passivization* are being identified as hedges; it does not seem to be on the basis of epistemic modality. The writer is not displaying a lack of confidence in his own proposition but politeness towards the discourse community (in this case, language testing researchers) whose face is being threatened by his observations. Again it seems that *some* politeness strategies are being included as hedges without any statement of a guiding principle.

The "IF-clauses" and "time reference" on Hyland's list of forms of hedge also appear questionable. Consider the following instance:

> If we eliminated the items missed by more than 10% of the EFL speakers, we would have only 14 out of 50 items left in the test. (Lado 1986: 133)

Clearly, the "IF-clause" modifies the main clause, but to describe it as a hedge or "indication of tentativeness" seems inappropriate. Again, there is no question of the writer displaying a lack of confidence in his proposition; rather the IF-clause is an essential part of the proposition. "IF-clauses" and "time reference" unquestionably *qualify* cognition; whether this is the same as saying that they *hedge* it seems debatable.

Part of the problem here seems to be a certain crudity in applying the concept of epistemic modality to real data. Lyons (1981), who developed the concept, writes of "the different resources, prosodic, grammatical and lexical, by which a locutionary agent can qualify his epistemic commitment to his proposition" (p. 238). By "epistemic verbs" Salager-Meyer presumably intends verbs which express less than absolute epistemic commitment, such as her examples "to suggest", "to speculate".

In fact, epistemic modality can only be a property of sentences, not of words or word classes. Failure to recognise this has meant that an important question has been overlooked: Is a writer who reports another responsible for the first's qualifications? To use a real example, if Carroll in 1972 wrote "I suggest that *p*", or "Perhaps *p*" (where *p*=a proposition), he was clearly hedging. But if Jafarpur in 1987 wrote "Carroll (1972) suggests that *p*", was Jafarpur (1987) really hedging? To judge from their examples, both Salager-

Meyer and Hyland would answer "yes"; I would prefer, however, to answer "no" and argue that Jafarpur's sentence is a new proposition, objective, historical, and falsifiable, different from *p*, with no personal tentativeness or epistemic qualification at all.

As Prince et al. (1982: 89) noted in discussing *attribution shields*, "the speaker's own degree of commitment" is "only indirectly inferrable". Skelton (1988b: 101), too, notes that there is a problem in distinguishing whether a reporting verb represents a comment (i.e. hedge) or merely a report.

To count all uses of certain linguistic tokens as hedges, is to run the risk of misrepresenting the discourse. We could for example designate the word "believe" as a member of the class of "epistemic verbs" (Salager-Meyer 1994) or, more accurately, "lexical verbs... expressing epistemic modality" (Hyland 1994). However, it is possible to imagine, to take an extreme example, a history of religious belief written without a single expression of tentativeness on the author's part ("The Egyptians believed that people had immortal souls", etc.). A mere tally of so-called epistemic verbs would, however, result in the work being characterized as high in epistemic modality, or densely hedged. What seems to be in danger of being overlooked in the lexical approach to quantifying epistemic modality, therefore, is the issue of *responsibility* for propositions. I will return to this issue later in attempting my own definition of *hedge*.

If, for the moment, however, we override such objections and include all lexical verbs reporting other peoples' propositions as hedges, we ought logically to include, as Hyland does but neither Skelton nor Salager-Meyer do, related items in other word classes, such as "assumption", "conclusion", "assertion", "supposed", and "purportedly". Perhaps the reason for the omission is that even more than reporting verbs such items clearly highlight the distance between any proposition and the author reporting it. For example, "Smith's *claim* that the moon was made of cheese was disputed by Jones..." seems even less like a hedged proposition than "Smith *claimed* that the moon was made of cheese. However, Jones...".

Salager-Meyer's (1994: 155) taxonomy of hedges has four main categories, summarized below:

1. **shields**: modal verbs expressing possibility, semi-auxiliaries (*appear*), probability adverbs (*probably*) and their derivative adjectives, epistemic verbs (*suggest*);
2. **approximators**: (*roughly, somewhat, often*);
3. **expressions of the authors' personal doubt and direct involvement**: (*we believe*);
4. **emotionally charged intensifiers**: (*particularly encouraging*).

Categories (3) and (4) cover politeness strategies as discussed earlier. Categories (1) and (2) are attempts to characterize formally the functional distinction of Prince et al. (1982). Salager-Meyer includes *approximators* in her taxonomy while acknowledging that they would not always count as hedging according to the ordinary use of *hedge*—"not all approximators

TABLE 1
Categories of Hedging Devices Recognized by Two or More Researchers

Hedging device	Skelton (1988)	Myers (1989)	Salager-Meyer (1994)	Hyland (1994)
• copulas other than be	✓	✓	✓	✓
• lexical verbs	✓(comment)	✓	✓(epistemic)	✓(epistemic)
• modal verbs	✓(all)	✓(making a conditional statement)	✓(expressing possibility)	✓(epistemic)
• probability adverbs	—	✓	✓	✓
• probability adjectives	—	—	✓	✓

serve to make things vague" (p. 1994: 154). However, the theoretical basis for their inclusion appears questionable:

> Approximators is the hedging category which most closely reflects what we would call the 'institutionalised' language of science.

Here, the scope of *hedge* seems to be being expanded to embrace yet another concept—"scientific language"—which has itself not been clearly defined.

Hedging: A Consensus?

This brief study of how the term *hedge* has been used in the literature, summarized in Tables 1 and 2, suggests then that there has, in practice, been little consensus on what the term *hedge* denotes.

The only item on which there seems to be complete agreement is copulas other than *be;* the extent and nature of modal and lexical verbs counted varies between scholars. Some kind of consensus clearly needs formulating

TABLE 2
Categories of Hedging Device Recognized by Only One Researcher

Skelton (1988b)	Myers (1989)	Salager-Meyer (1994)	Hyland (1994)
• all clause initial adverbs	• all devices suggesting an alternative	• lexis expressing personal involvement	• IF clauses
• all adjectives in introductory phrases		• emotionally charged intensifiers	• time adverbials
		• approximators	• impersonal expressions
			• passives
			• modal nouns, adjectives and adverbials (other than probability)

if ESP materials writers and teachers are indeed to set about raising students' awareness of hedging in academic discourse.

It seems clear that hedging cannot, unfortunately, be pinned down and labelled as a closed set of lexical items. The only possible set of identifiable items (e.g. "approximately", "somewhat", "quite") would be *approximators* (Prince et al. 1982). However, it is noteworthy that Hyland (1994), Skelton (1988b), and Myers (1989) all leave this category out of their taxonomies altogether. This is at first sight odd because here is a class of words which most obviously seems to meet G. Lakoff's original definition (1972: 195) of hedges as "words which make things fuzzy or less fuzzy". I can only speculate that the reason for this omission is that these researchers have each attempted to look beyond purely formal criteria and apply a single functional purpose to hedging; it seems clear that as a class *approximators* does not fit in with those purposes (characterized as *comment*, *provisionality*, or *epistemic modality* respectively). "Fuzziness" per se seems too fuzzy or general a concept to serve as a basis for a useful definition of *hedge*. The correctness of Prince et al.'s original *approximator/shield* distinction seems vindicated (*pace* Skelton 1988a: 38). The fuzziness which seems to be likely to be of most value to students of academic writing is the pragmatic fuzziness of "marked commitment" (Prince et al. 1982: 86) relating to the relationship between writer and proposition (*shields*) rather than the semantic fuzziness relating to lexis within propositional content (*approximators*).

A Proposed Definition of Hedging

I agree with the implicit suggestion of the researchers mentioned above that a functionally-based definition of hedging is desirable. Without such a definition, the term designates a "rag-bag" category of features noticed in academic/scientific writing, understood by different people in different ways. If a coherent functional description of hedging is to be established, the case for tying it to epistemic modality seems strong. However, as the earlier discussion on the role of hedging in politeness theory suggests, it should be clear that the issues involved are pragmatic and rhetorical as well as cognitive. Hedging may be used to display not only or necessarily the degree of confidence speakers have in their propositions but also how much confidence they feel it is appropriate to display.

Borrowing Lyons's original definition (1977: 797) of epistemic modality I would, then, like to suggest the following definition for hedges in academic writing. The definition applies only to hedges on propositions, the main kind of speech act performed in academic writing:

> A hedge is an item of language which a speaker uses to explicitly qualify his/her lack of commitment to the truth of a proposition he/she utters.

I would further like to propose the following simple test for determining whether or not a proposition is hedged:

> Can the proposition be restated in such a way that it is not changed but that the author's commitment to it is greater than at present? If "yes" then the proposition is hedged. (The *hedges* are any language items in the original which would need to be changed to increase commitment.)

This test addresses the issues of (a) whether or not inexact language (*approximators*) is being used to diminish commitment or merely as shorthand, and (b) who is responsible for any tentativeness in stating propositions (i.e. the authors or the sources they are citing). Incidentally, the test also helps to distinguish whether instances of modal verbs carry an epistemic or root meaning (Butler 1990: 145–147).

Let us apply this test to some real data, all cited by Salager-Meyer as typical exponents of hedging in her medical research corpus:

1. *Little information* exists on the frequency and severity of the disorder.
2. Ciguatera poisoning is *usually* a clinical diagnosis.

It is difficult to see how either of these could be altered with the effect of increasing the author's commitment. To remove, or change the "hedges" actually changes the propositional content, e.g.

1a. Information exists on the frequency and severity of the disorder.
1b. *No* information exists on the frequency and severity of the disorder....
2a. Ciguatera poisoning is a clinical diagnosis
2b. Ciguatera poisoning is *always* a clinical diagnosis.

In terms of the proposed test, then "little" and "usually" would both fail to qualify as instances of hedging. Let us contrast this with the following:

3. I'm *rather* hungry.

Removing "rather" increases the speaker's commitment but doesn't alter the basic proposition:

3a. I'm hungry.

The above instance of "rather" would pass the test as a hedge, then. In practice, this kind of hedge ("somewhat", "sort of", "-ish")—*adaptors* (Prince et al. 1982)—though frequent in the spoken corpus of Prince et al. seems difficult to find in academic writing. The test disqualifies most of the approximators found in the "Methods" sections of Salager-Meyer's corpus (1994) of medical research articles—*rounders* (Prince et al. 1982), e.g. "*about* 5 hours", "*roughly* the same as" (Salager-Meyer 1994: 161). Prince et al. admit that this class of items "do not reflect uncertainty or fuzziness but are rather a shorthand device when exact figures are not relevant or available" (p. 95).

Turning to the responsibility issue, let us consider the following, also from Salager-Meyer's corpus:

1. Shmerling *suggested* that sensitization took place during the first hours after birth.

2. Previous estimates of the incidence of acute mountain sickness *suggest* that...

In (1), one has to, in Gricean terms, assume the truth of "Shmerling suggested" and that a verb denoting a greater or lesser level of epistemic commitment on Shmerling's part would be dishonest on the part of the author. It might possibly be argued, however, that the proposition being hedged is the proposition actually being reported; on this view the author could have avoided the hedge by writing simply:

1a. Sensitization takes place during the first hours after birth.

or in reporting Shmerling's proposition, evaluate it in such a way as to express total confidence, and hence commitment:

1b. Shmerling showed that sensitisation takes place during the first hours after birth.

On this view, the use of *non-factive* reporting verbs (Leech 1983), where the author does not commit him/herself for or against the reported proposition (e.g. "claim", "argue", "assert"), would therefore always be counted as hedging; only *factive* (e.g. "showed") and *counterfactive* (e.g. "pretended") reporting verbs, where authors commit themselves to either endorsement or denial of the reported proposition, would be regarded as *not* hedging. As I suggested earlier, however, I believe this approach risks misrepresenting the discourse by failing to distinguish between authors' own propositions and those they attribute to others, which they may wish to discuss, and will quite possibly go on to contradict. The test I propose, therefore, attempts to make such a distinction by addressing the issue of responsibility for utterance. The use of any kind of reporting verb only counts as a hedge if authors have elected to use them to report their own proposition; thus, for example, "I suggest that pigs fly" would be regarded as a hedged version of "Pigs fly", whereas "Smith suggests that pigs fly" would not.

According to the test, then, (1) mentioned earlier ("Shmerling suggested...") would not count as a hedge because the author is not responsible for the reported proposition. However, with (2) one could strengthen the level of commitment of "suggest" because this verb represents the *author's* interpretation of the "previous estimates". The author could, then, show greater commitment to his/her interpretation of the subsequent proposition without altering the proposition per se: e.g.

2a. Previous estimates of the incidence of acute mountain sickness *show* that...

The fact that the author has chosen the verb "suggest" denotes an explicit display of lack of commitment; according to the test, therefore, (2) would be designated as hedged, with "suggest" as the hedging device or hedge.

A Proposed Taxonomy of Hedges

Turning to a taxonomy of hedges, an identification of hedges as individual words seems inappropriate (although the identification of certain commonly

employed words might be a starting point for corpus-based studies of the phenomenon). A more useful approach might be to compile a list of common sentence patterns. As a start, and based largely on the common core of uncertainty as it has featured in the research work already discussed, I suggest the following characterisations of hedged propositions:

1. Sentences with copulas other than *be.*
2. Sentences with modals used epistemically.
3. Sentences with clauses relating to the probability of the subsequent proposition being true.
4. Sentences containing sentence adverbials which relate to the probability of the proposition being true.
5. Sentences containing reported propositions where the author(s) can be taken to be responsible for any tentativeness in the verbal group, or non-use of factive reporting verbs such as "show", "demonstrate", "prove". These fall into two sub-types:
   a. where authors explicitly designate themselves as responsible for the proposition being reported;
   b. where authors use an impersonal subject but the agent is intended to be understood as themselves.
6. Sentences containing a reported proposition that a hypothesized entity X exists and the author(s) can be taken to be responsible for making the hypothesis.

The following constructed examples illustrate the basic kinds of hedge in each type of hedged proposition:

1. The moon *appears to be* made of cheese.
2. The moon *might be* made cheese.
3. *It is likely that* the moon is made of cheese.
4. The moon is *probably* made of cheese.

5a. *I suggest that* the moon is made of cheese.

5b. It is therefore *suggested that* the moon is made of cheese.

6. These findings *suggest* a cheese moon.

Compounding of hedges is quite common, but the elements of each compound are still distinguishable: e.g.

These results *would seem to suggest that* the moon is made of cheese. (Types 2, 1, and 5b respectively).

Note that according to the proposed definition none of the following would count as hedged propositions:

Moons are *usually* made of cheese. (*Approximator.*)
The moon is made *mostly* of cheese.(*Approximator.*)
Smith (1996) *suggests that* the moon is made of cheese. (*Attribution shield*, "epistemic verb".)
It *has* commonly *been assumed that* the moon is made of cheese. (Impersonal construction, "epistemic verb".)

If Smith's (1996) *findings are accurate*, the moon is made of cheese. (IF-clause.)
On clear nights, the moon is made of cheese. (Time reference.)
Encouragingly, the moon is made of cheese. (Lexis suggesting authors' personal involvement.)
It has been shown (Smith 1996) *that* the moon is made of cheese. (Passive, impersonal construction.)

Interestingly, if "epistemically modal" verbs are being used to perform "speech acts" they cannot be counted as epistemically modal. The following instance of "suggest", for example, is not part of a proposition but of a proposal.

> We would finally like to suggest at least one line of future research: (Salager-Meyer 1994: 166)

It would not be meaningful to discuss such a sentence in terms of the speaker's commitment. Note also that although the sentence refers to a hypothesized entity it does not contain a proposition relating to that entity's existence or non-existence; it is not, therefore, a type 6 hedge as described above.

Myers (1989): 17) claims that type 5b is the most common form for a statement of a knowledge claim. The verb "suggest" is common: e.g.

> In sum, this survey suggests that in the area of academic hedging, the pedagogy of ESP needs revision. (Hyland 1994: 253)

However, as well as "suggest", "imply" and "indicate", other more complex verbal groups may be used, e.g. "this leads to the proposal that", "supports the position that", "points to the fact that". Note that what qualifies these expressions as hedges is not their impersonal subjects, but the fact that the author could have expressed complete epistemic commitment (e.g. "demonstrates that", "confirms that") but has chosen not to. Instead of the above, for example, Hyland could have written:

> In sum, this survey *shows* that in the area of academic hedging, the pedagogy of ESP needs revision.

Type 6 is closely related to type 5; the following are authentic examples (cited in Myers 1989):

1. These results imply a novel mechanism for biosynthesis of SV40 mRNA.
2. These findings suggest a common origin of some nuclear and mitochondrial istrons in the mechanism of their splicing.

With regard to the test I have proposed, it does not seem possible, however,

to strengthen the commitment of these sentences simply by uprating "suggest" or "imply" to "show" or "prove":

1a. These results *show* a novel mechanism for biosynthesis of SV40 mRNA.

The phrases "imply an X" and "suggest a Y" seem to be paraphrasable as "imply that X exists" and "suggests that Y exists" respectively. To strengthen the epistemic commitment of these sentences one would need to restructure the sentence and change the verb to produce something like the following:

1b. These results *show* that there is a novel mechanism for biosynthesis of SV40 mRNA.
2b. These findings *show* that some nuclear and mitochondrial istrons have a common origin in the mechanism of their splicing.

If it is accepted that it is propositions rather than sentences or sub-sentential groups which are hedged (or epistemically modalized), there should be no objection to restructuring of this kind.

Conclusion

It seems that there is a danger of *hedge* being used as a catch-all term for an assortment of features noticed in academic writing. Clearly, the use of impersonal constructions, passivization, lexis expressing personal involvement, other politeness strategies, and factivity in reporting/evaluating the claims of other researchers are important issues in academic writing; these all seem worthy of further research to enhance the teaching of the subject. However, the restriction of *hedge* to designate language avoiding commitment, a use which corresponds closely, as we have seen, with the ordinary use of the word, seems desirable and feasible, both theoretically and pedagogically. Equipped with such a functional definition, it should be easier, both for teachers and for students, to identify and talk about the major kinds of hedge to be found in the target discourse.

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